DEVOPS

LAB 02

Hands-On Git and Version Control

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Registration Number: 2023-BSCS-33

Course Code: BCS-649

Section: A

GitHub Username: fatimanoorulimran

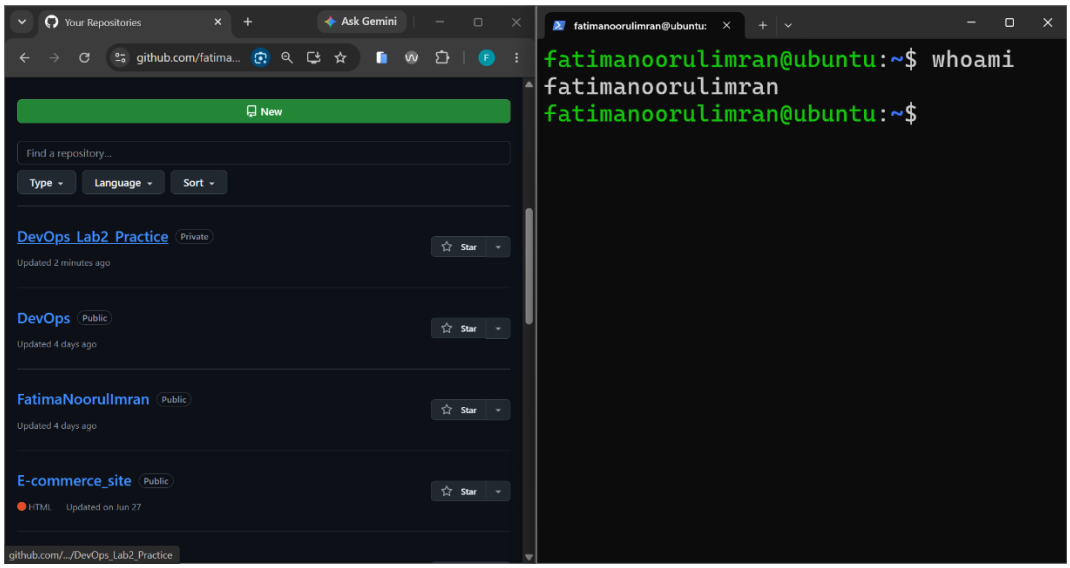
Getting Started: Install Git

```
fatimanoorulimran@ubuntu:~$ sudo apt install -y git
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30                linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-headers-7.0.0-30-generic        linux-modules-7.0.0-30-generic
  linux-image-unsigned-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 31
fatimanoorulimran@ubuntu:~$ git --version
git version 2.53.0
fatimanoorulimran@ubuntu:~$
```

Git installed on Ubuntu, with git --version output and my prompt fatimanoorulimran@ubuntu.

Task 1: Create Private GitHub Repository

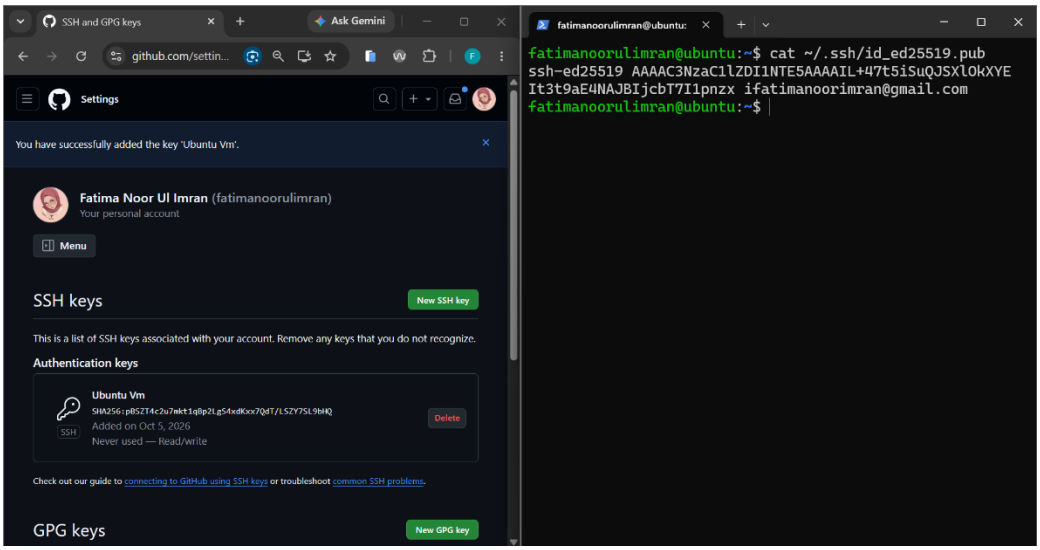


Repository DevOps_Lab2_Practice marked Private on GitHub, next to my terminal prompt.

Task 2: Connect Repository via SSH

```
fatimanoorulimran@ubuntu:~$ ssh-keygen -t ed25519 -C "ifatimanoorimran@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/fatimanoorulimran/.ssh/id_ed25519):
Enter passphrase for "/home/fatimanoorulimran/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/fatimanoorulimran/.ssh/id_ed25519
Your public key has been saved in /home/fatimanoorulimran/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:pBSZT4c2u7mkt1qBp2LgS4xdKxx7QdT/LSZY7SL9bHQ ifatimanoorimran@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|     ...o .      |
|    .  +.= .     |
|     . .=.=      |
|    . . += .     |
|   o o o+S= .    |
|  * = oooB.* E   |
| . O = .+.O o    |
|  . = ...o +     |
|   .    .o.o     |
+----[SHA256]-----+
fatimanoorulimran@ubuntu:~$ cat ~/.ssh/id_ed25519.pub
```

Generating the ed25519 SSH key and displaying only the public key.



Public SSH key added to GitHub under Settings > SSH and GPG keys.

```
fatimanoorulimran@ubuntu:~$ git clone git@github.com:fatimanoorulimran/DevOps_Lab2_Practice.git
Cloning into 'DevOps_Lab2_Practice'...
warning: You appear to have cloned an empty repository.
fatimanoorulimran@ubuntu:~$ ls
DevOps_Lab2_Practice
fatimanoorulimran@ubuntu:~$
```

Repository DevOps_Lab2_Practice cloned over SSH.

Task 3: Configure Git Username and Email

```
fatimanoorulimran@ubuntu:~$ git config --global user.name "Fatima Noor ul Imran"
fatimanoorulimran@ubuntu:~$ git config --global user.email "ifatimanoorimran@gmail.com"
fatimanoorulimran@ubuntu:~$
fatimanoorulimran@ubuntu:~$
fatimanoorulimran@ubuntu:~$ git config --global user.name
Fatima Noor ul Imran
fatimanoorulimran@ubuntu:~$ git config --global user.email
ifatimanoorimran@gmail.com
fatimanoorulimran@ubuntu:~$
```

Git user.name and user.email configured.

```
fatimanoorulimran@ubuntu:~$ git config --list
user.name=Fatima Noor ul Imran
user.email=ifatimanoorimran@gmail.com
fatimanoorulimran@ubuntu:~$
```

git config --list output showing my identity.

Task 4: Explore the .git Folder

```
fatimanoorulimran@ubuntu:~$ cd DevOps_Lab2_Practice/
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ ls -a
.  ..  .git
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ ls -a .git
.  ..  HEAD  config  description  hooks  info  objects  refs
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

Hidden .git directory and its contents listed with ls -a .git.

Task 5: Local Repository Management

```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ rm -rf .git
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git status ==> error why?
fatal: not a git repository (or any of the parent directories): .git
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

After rm -rf .git, git status reports that this is no longer a git repository.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/fatimanoorulimran/DevOps_Lab2_Practice/.git/
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

Repository re-initialized with git init.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ echo "# Lab02 Git Practice" >> README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Initial commit"
[master (root-commit) d21127b] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

README.md created and committed as the Initial commit.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git remote add origin git@github.com:fatimanoorulimra
n/DevOps_Lab2_Practice.git
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git branch -M main
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 243 bytes | 243.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:fatimanoorulimran/DevOps_Lab2_Practice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

Local repository connected to GitHub and pushed to main.

Task 6: File Status & Staging

```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ echo "Adding some test notes" >> notes.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        notes.txt

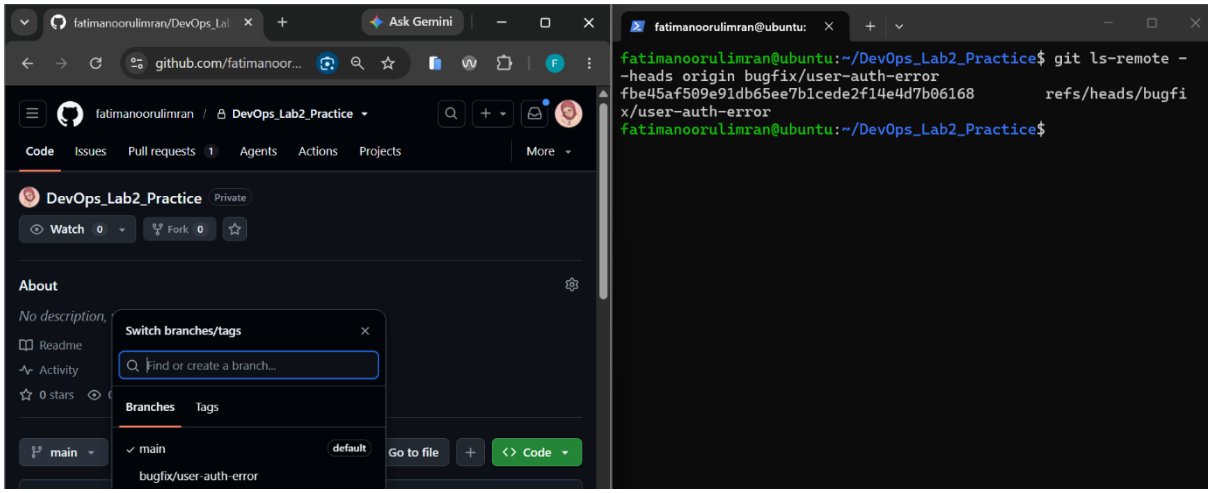
nothing added to commit but untracked files present (use "git add" to track)
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

git status showing notes.txt as an untracked file.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add notes.txt"
[main fbe45af] Add notes.txt
 1 file changed, 1 insertion(+)
 create mode 100644 notes.txt
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

notes.txt staged with git add and committed.

Task 7: Branch Creation Using GitHub GUI



Branch bugfix/user-auth-error created on GitHub, shown next to my terminal.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git pull origin bugfix/user-auth-error
From github.com:fatimanoorulimran/DevOps_Lab2_Practice
 * branch            bugfix/user-auth-error -> FETCH_HEAD
 * [new branch]      bugfix/user-auth-error -> origin/bugfix/user-auth-error
Already up to date.
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git fetch origin
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git checkout bugfix/user-auth-error
branch 'bugfix/user-auth-error' set up to track 'origin/bugfix/user-auth-error'.
Switched to a new branch 'bugfix/user-auth-error'
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git branch
* bugfix/user-auth-error
  main
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

Branch pulled and checked out locally, shown in the git branch list.

Task 8: Branch Creation and Push Using Git Bash

```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature/db-connection
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:      https://github.com/fatimanoorulimran/DevOps_Lab2_Practice/pull/new/feature/db-connection
remote:
To github.com:fatimanoorulimran/DevOps_Lab2_Practice.git
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

Branch feature/db-connection created locally and pushed to GitHub.

Task 9: Branching & Merging

```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature-1
Switched to a new branch 'feature-1'
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch feature-1
nothing to commit, working tree clean
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

Branch feature-1 created and checked out.

```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ cat main.py
def greet(name):
    return "Hello, " + name

print((greet("World"))
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git add main.py
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add new function to main.py"
[feature-1 b377d32] Add new function to main.py
 1 file changed, 4 insertions(+)
 create mode 100644 main.py
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

Function added to main.py and committed on feature-1.

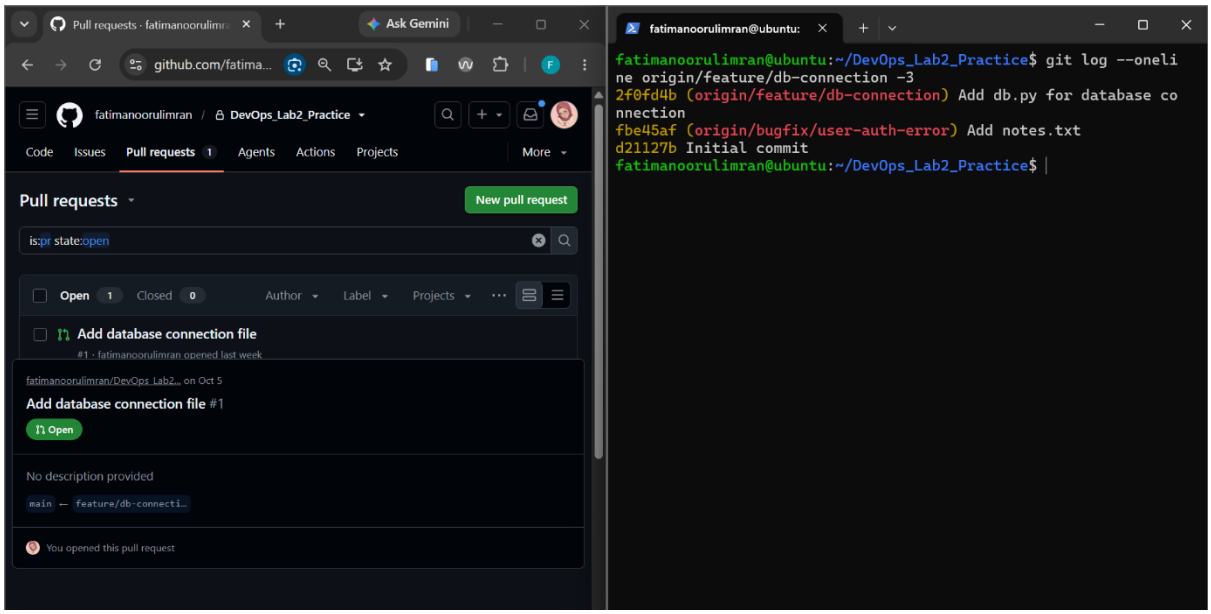
```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git merge feature-1
Updating fbe45af..b377d32
Fast-forward
 main.py | 4 ++++
 1 file changed, 4 insertions(+)
 create mode 100644 main.py
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

feature-1 merged into main.

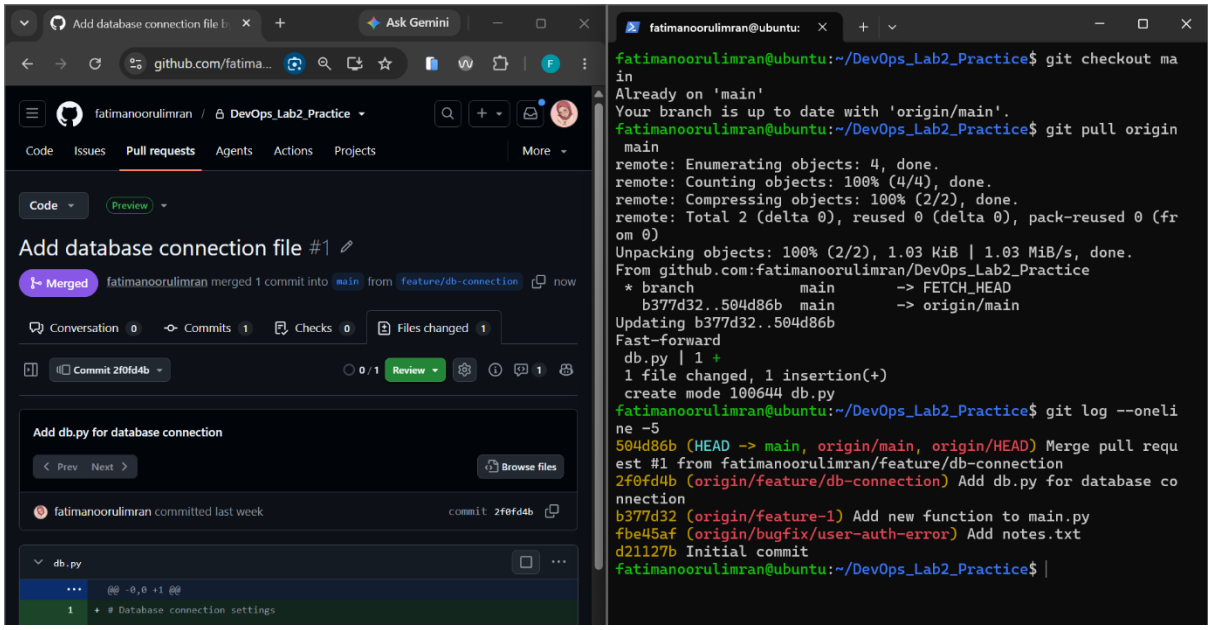
```
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 391 bytes | 130.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:fatimanoorulimran/DevOps_Lab2_Practice.git
   fbe45af..b377d32  main -> main
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature-1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:      https://github.com/fatimanoorulimran/DevOps_Lab2_Practice/pull/new/feature-1
remote:
To github.com:fatimanoorulimran/DevOps_Lab2_Practice.git
 * [new branch]      feature-1 -> feature-1
fatimanoorulimran@ubuntu:~/DevOps_Lab2_Practice$
```

Branches main and feature-1 pushed to GitHub.

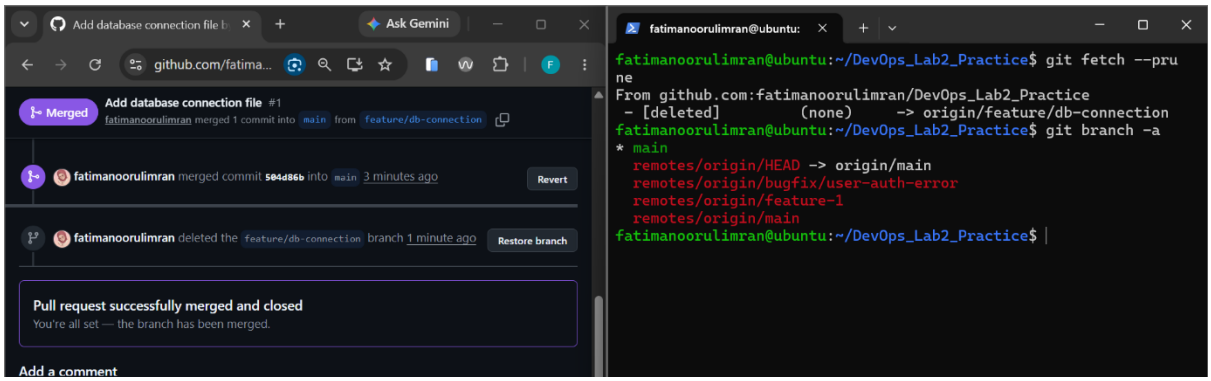
Task 10: Pull Request and Branch Review (GitHub GUI)



Pull request #1 opened from feature/db-connection into main.

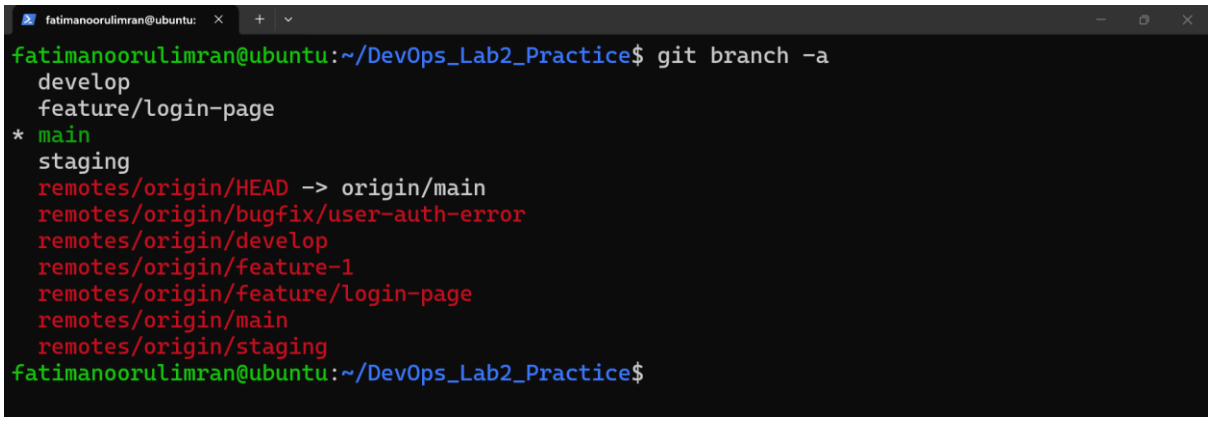


Pull request reviewed on the Files changed tab and merged using the GitHub interface.

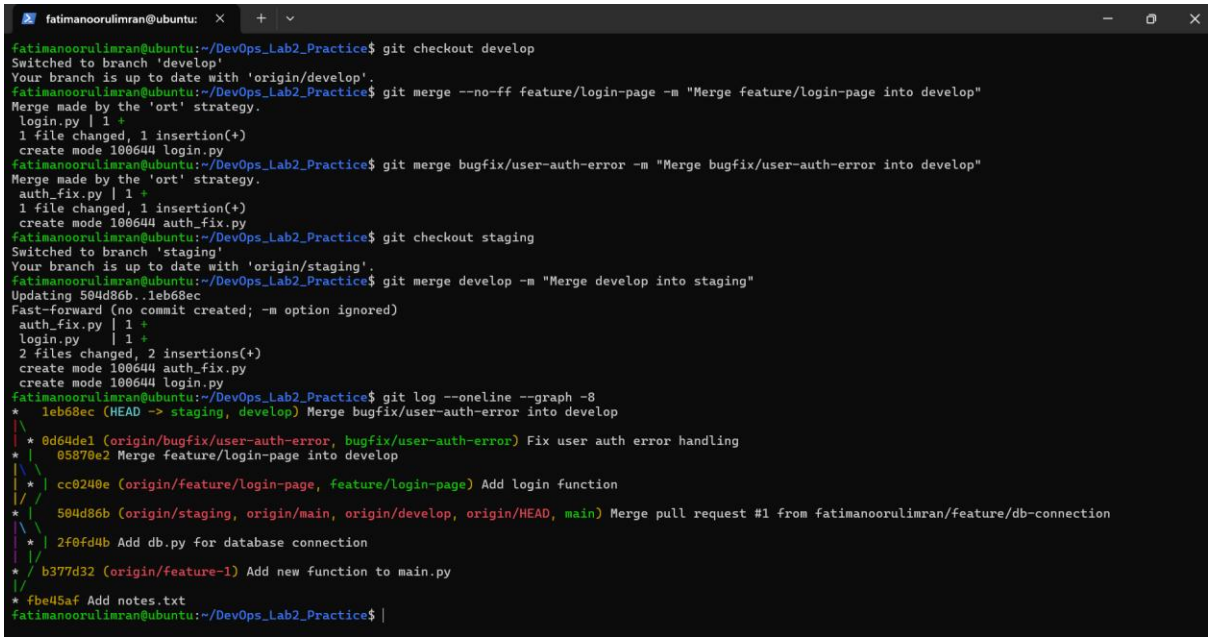


Branch feature/db-connection deleted after the merge.

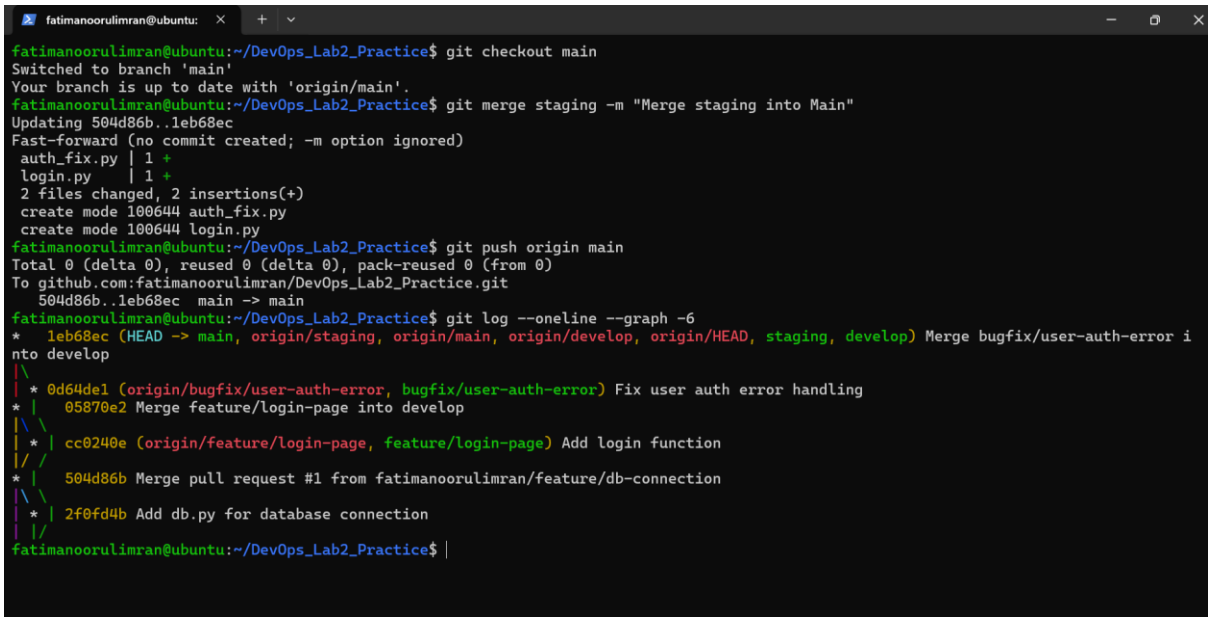
Task 11: Detailed Branch Strategy (Develop/Staging)



Branch list showing main, develop, staging, feature/* and bugfix/* branches.

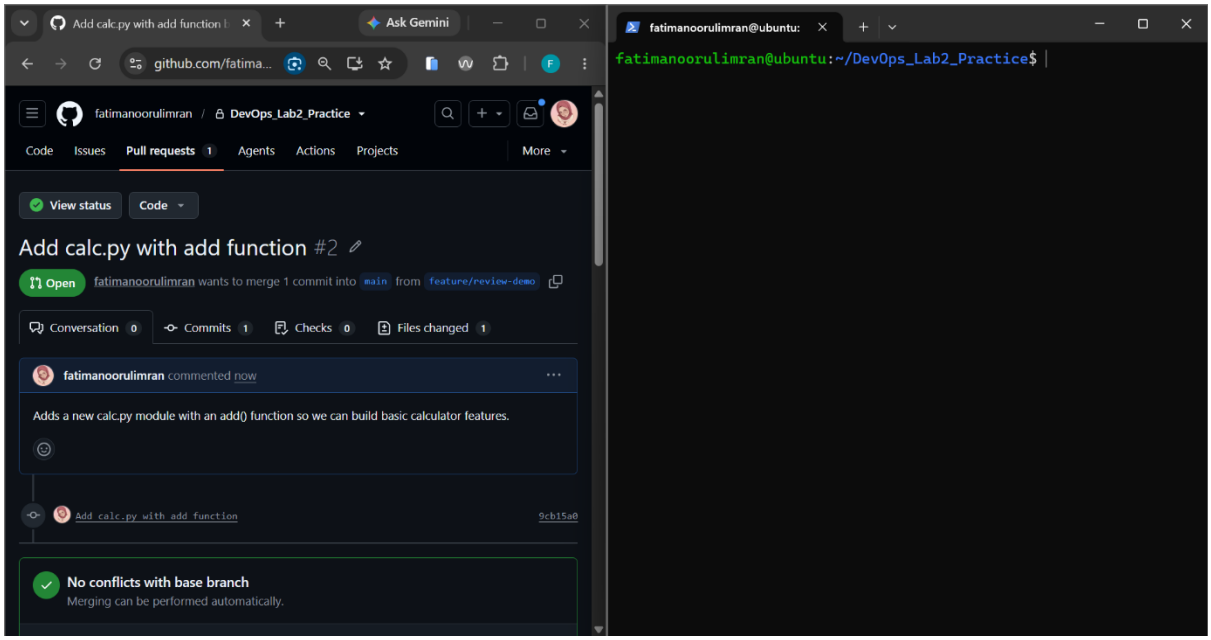


Feature and bugfix branches merged into develop, then develop merged into staging.

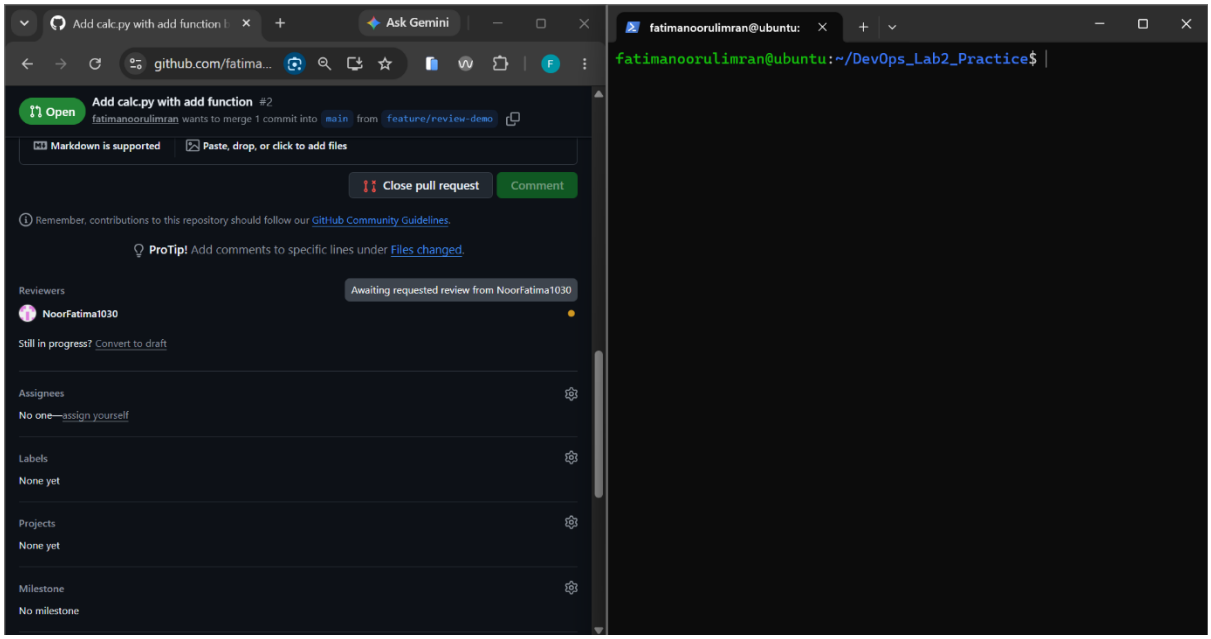


staging merged into main as the final production release.

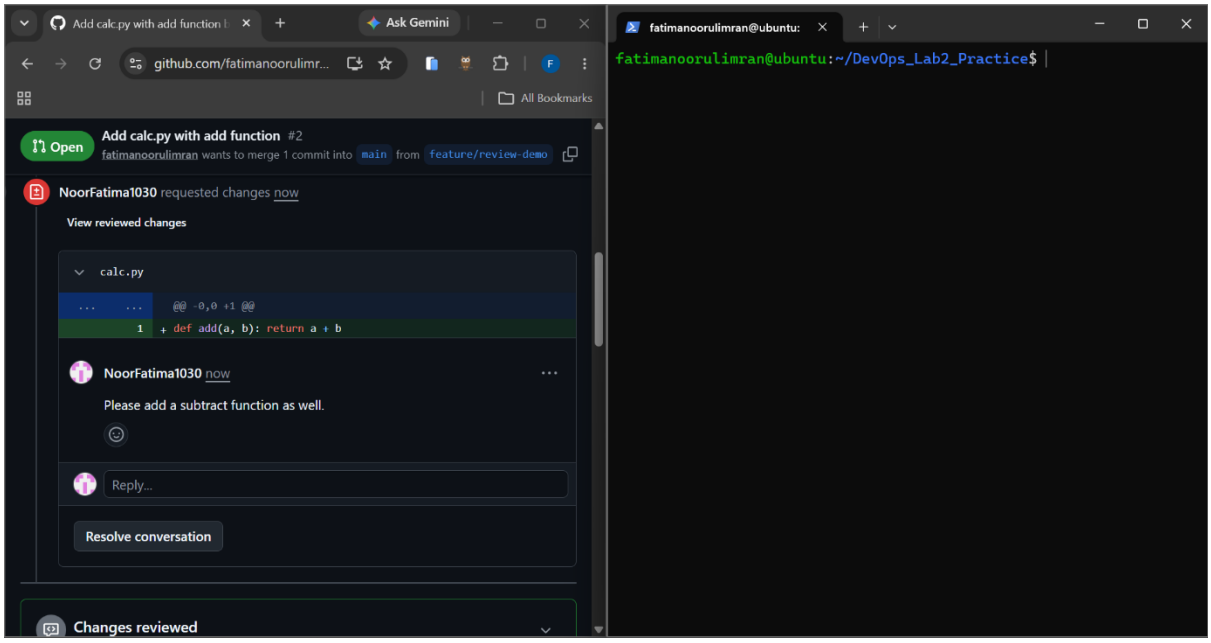
Task 12: Code Review Workflow



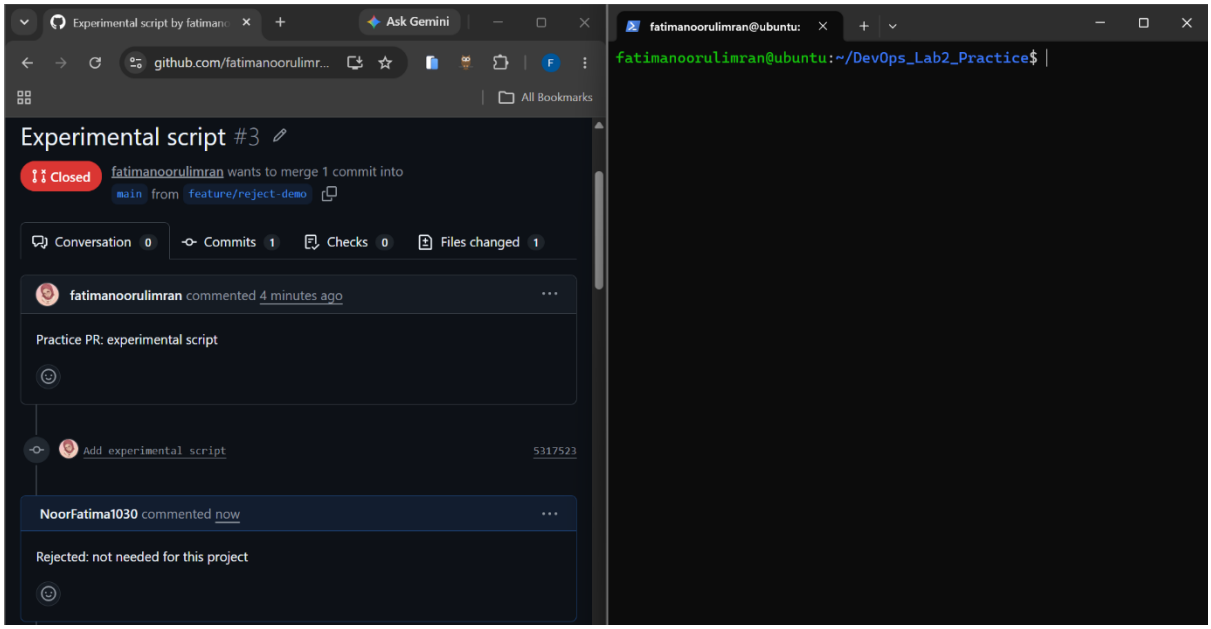
Pull request #2 with a clear title and description, from feature/review-demo into main.



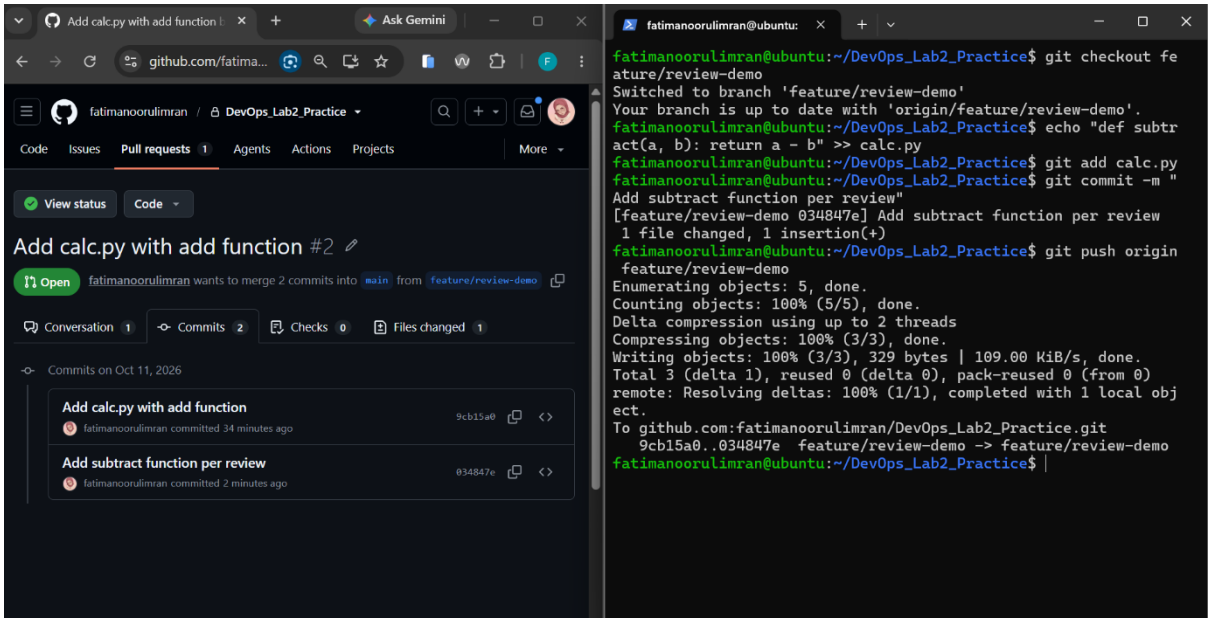
Second account NoorFatima1030 assigned as reviewer.



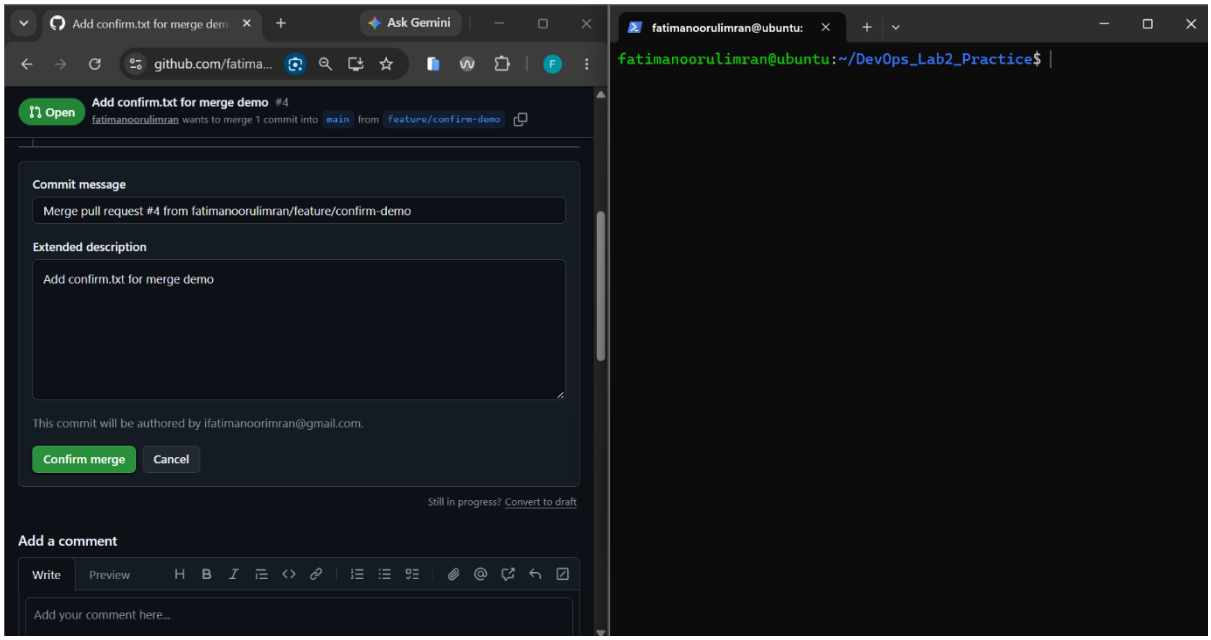
Reviewer requested changes and left a comment asking for a subtract function.



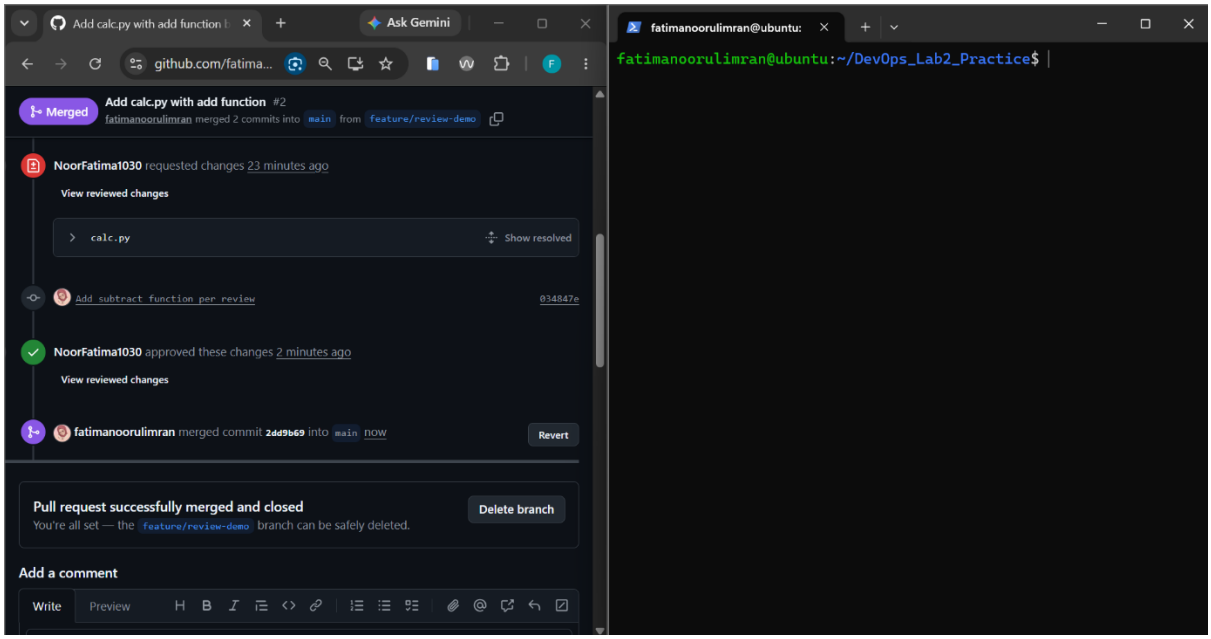
Practice pull request closed by the reviewer with the comment Rejected: not needed for this project.



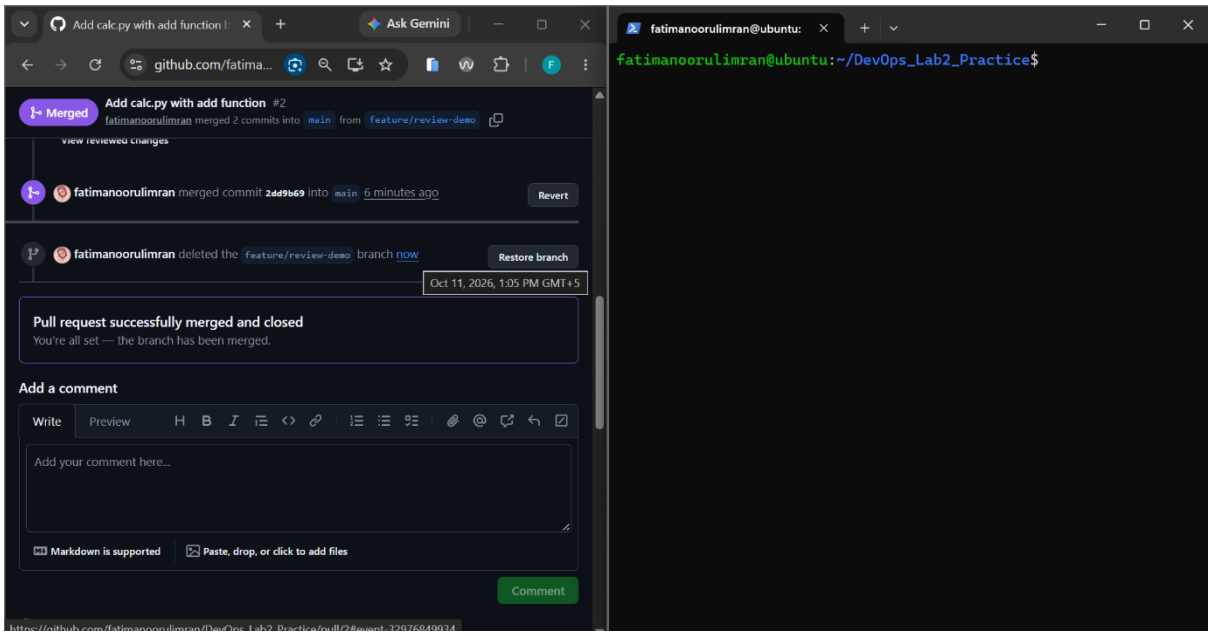
Fix pushed to the same branch; the pull request updated automatically with the new commit.



Merge confirmation screen before completing the merge.

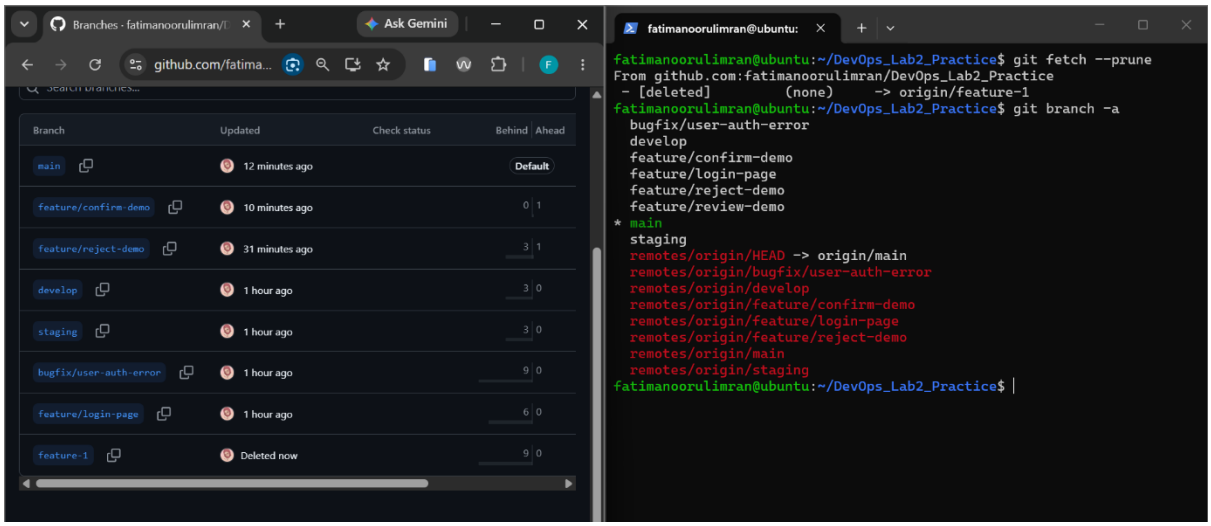


Pull request merged, shown with the Merged status.

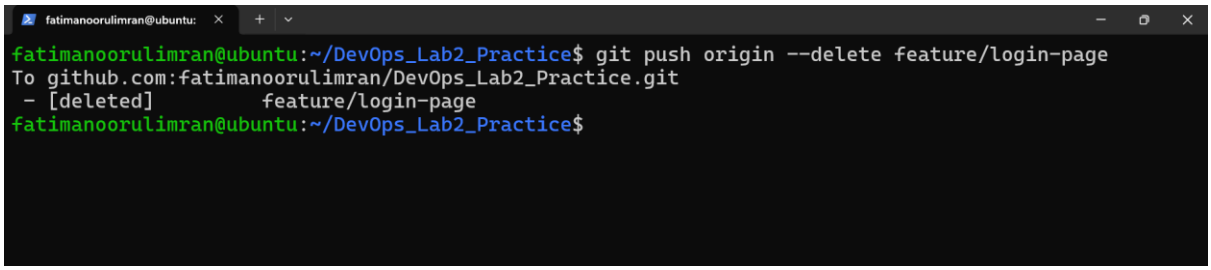


Merged branch deleted from GitHub.

Task 13: Branch Cleanup Best Practices

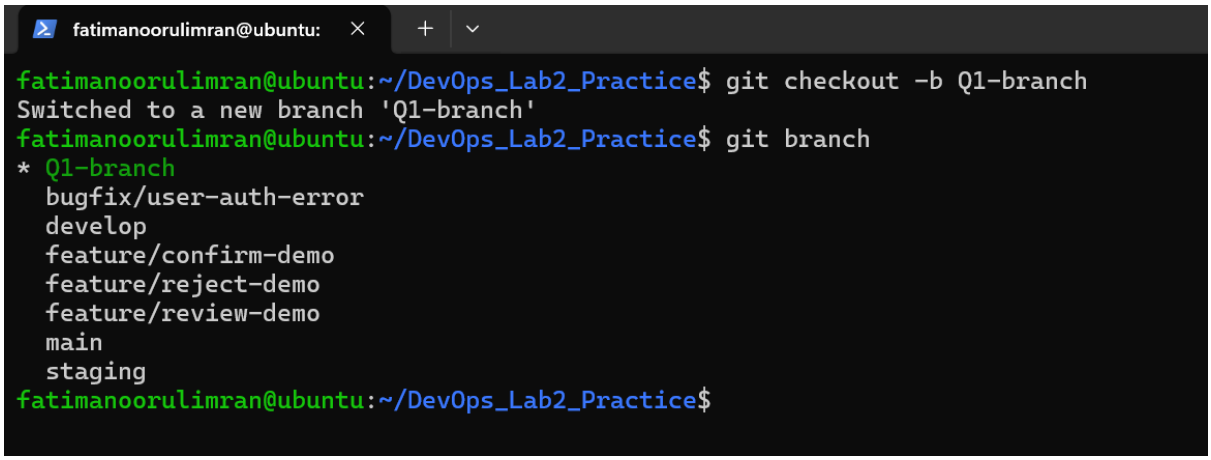


Merged remote branch deleted through the GitHub interface.

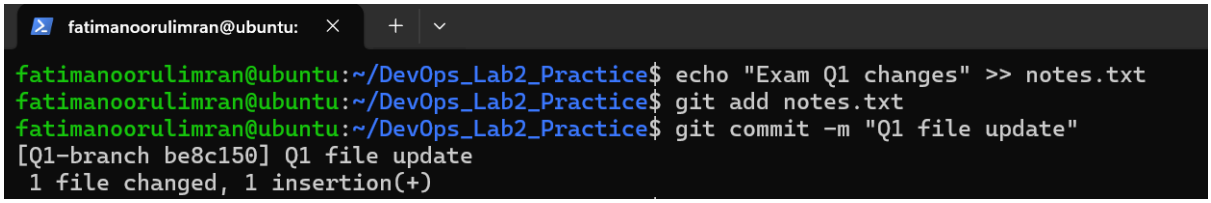


Remote branch deleted from the terminal with git push origin --delete.

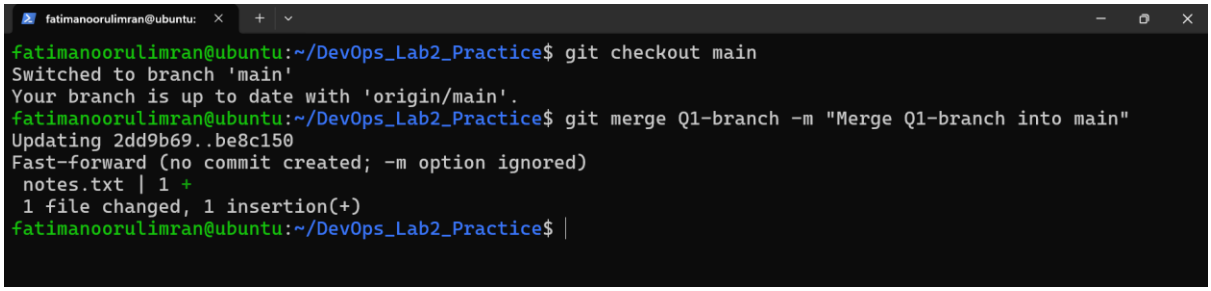
Exam Question 1: Advanced Branching & Merge Verification



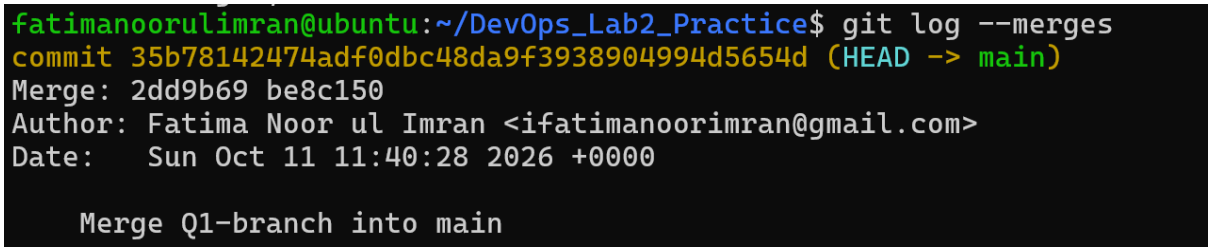
New branch created.



Small change made and committed on the new branch.

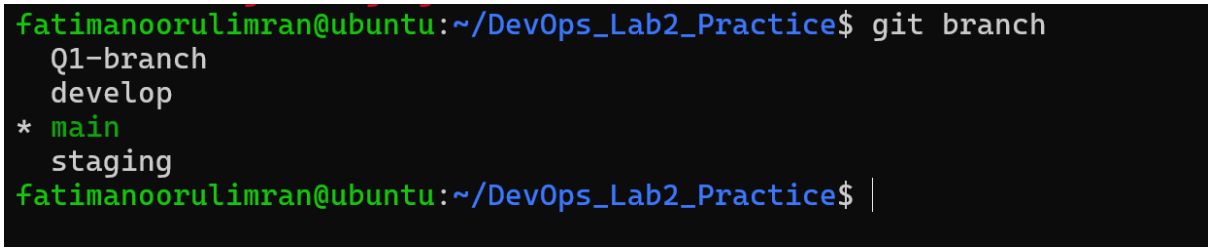


Branch merged back into main.

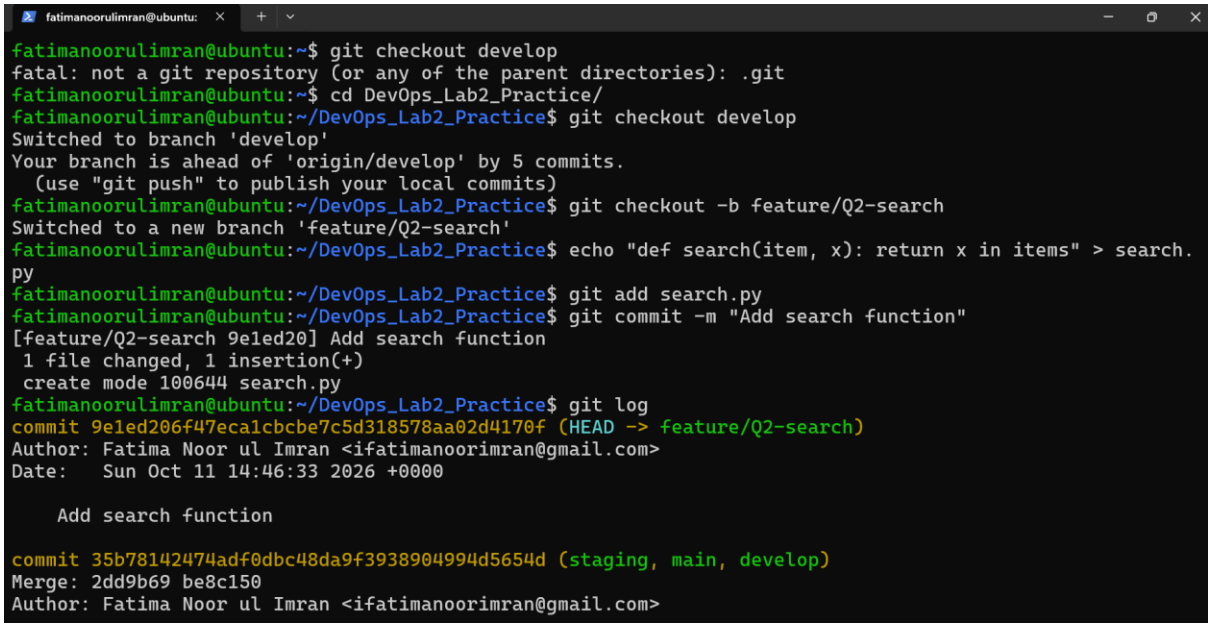


History log of merge history.

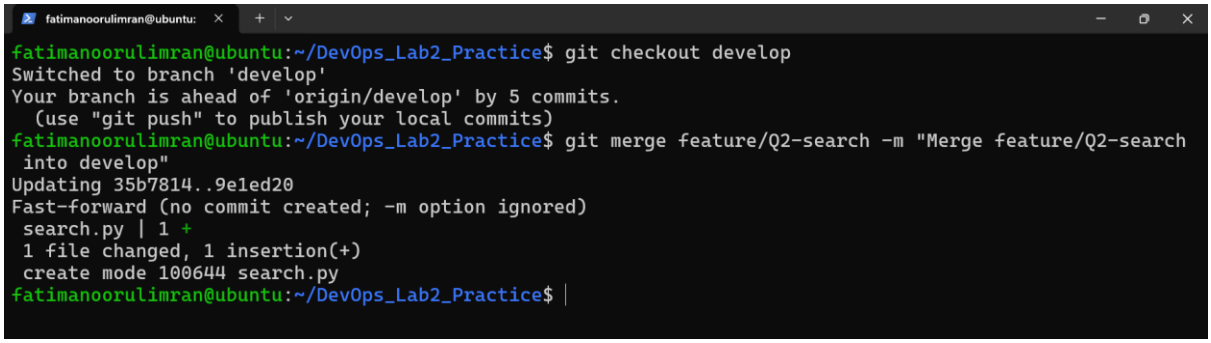
Exam Question 2: Multi-Stage Workflow Simulation



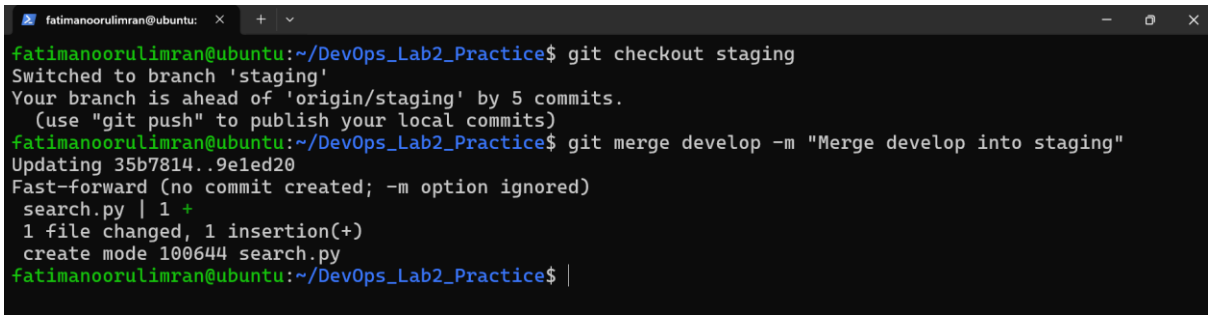
Branches main, develop and staging created.



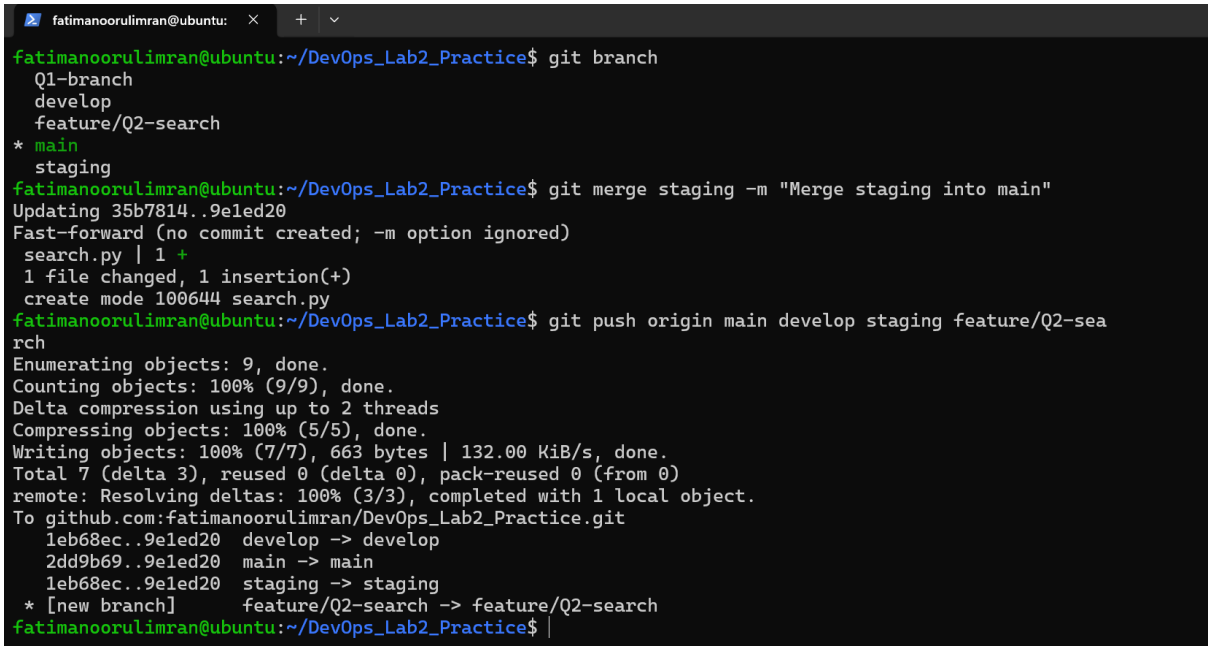
Feature branch created from develop with a commit.



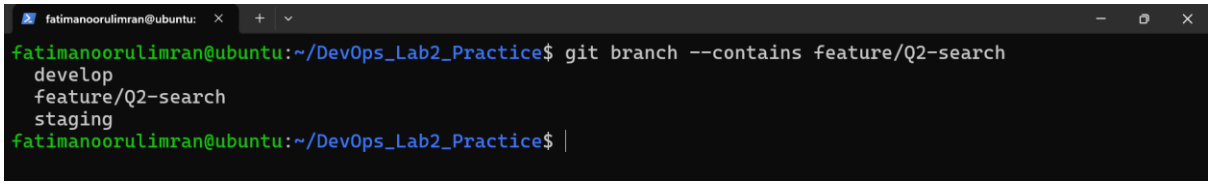
Feature branch merged into develop.



develop merged into staging.

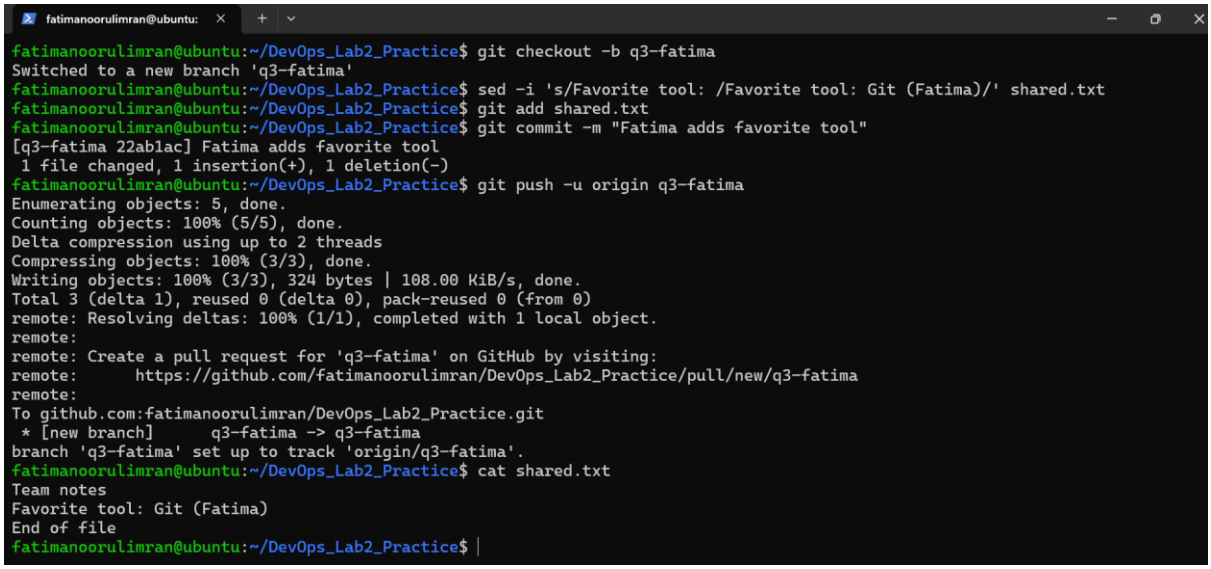


staging merged into main.

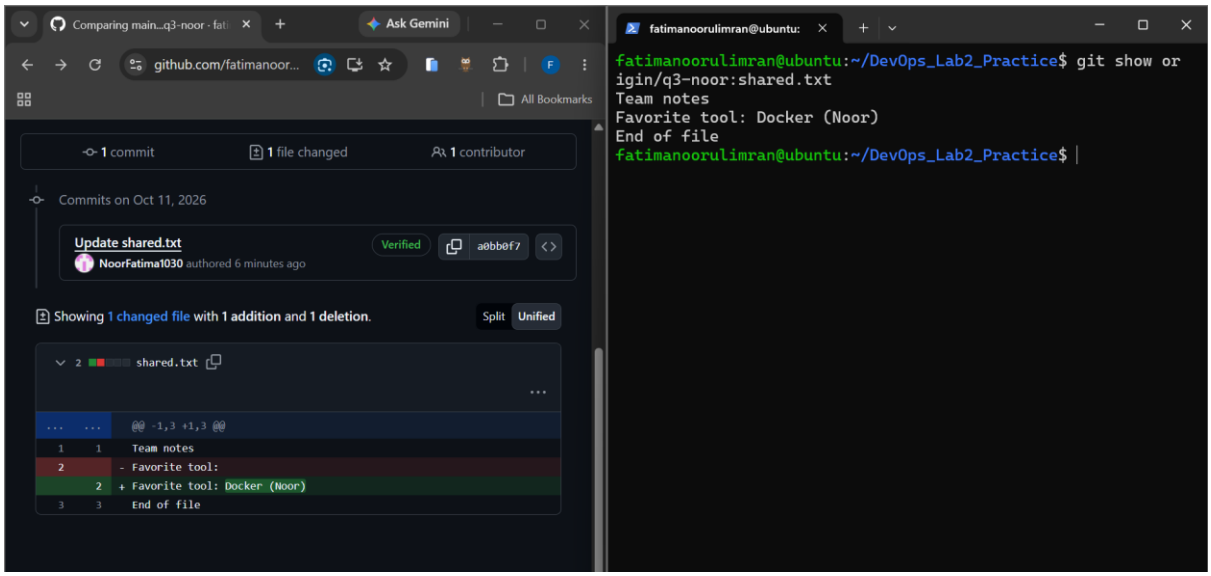


Proof that each stage contains the updated changes before main.

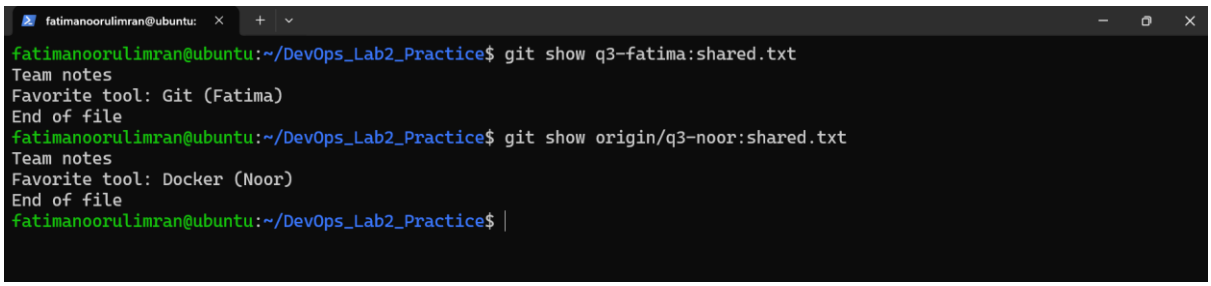
Exam Question 3: Collaboration & Conflict Resolution



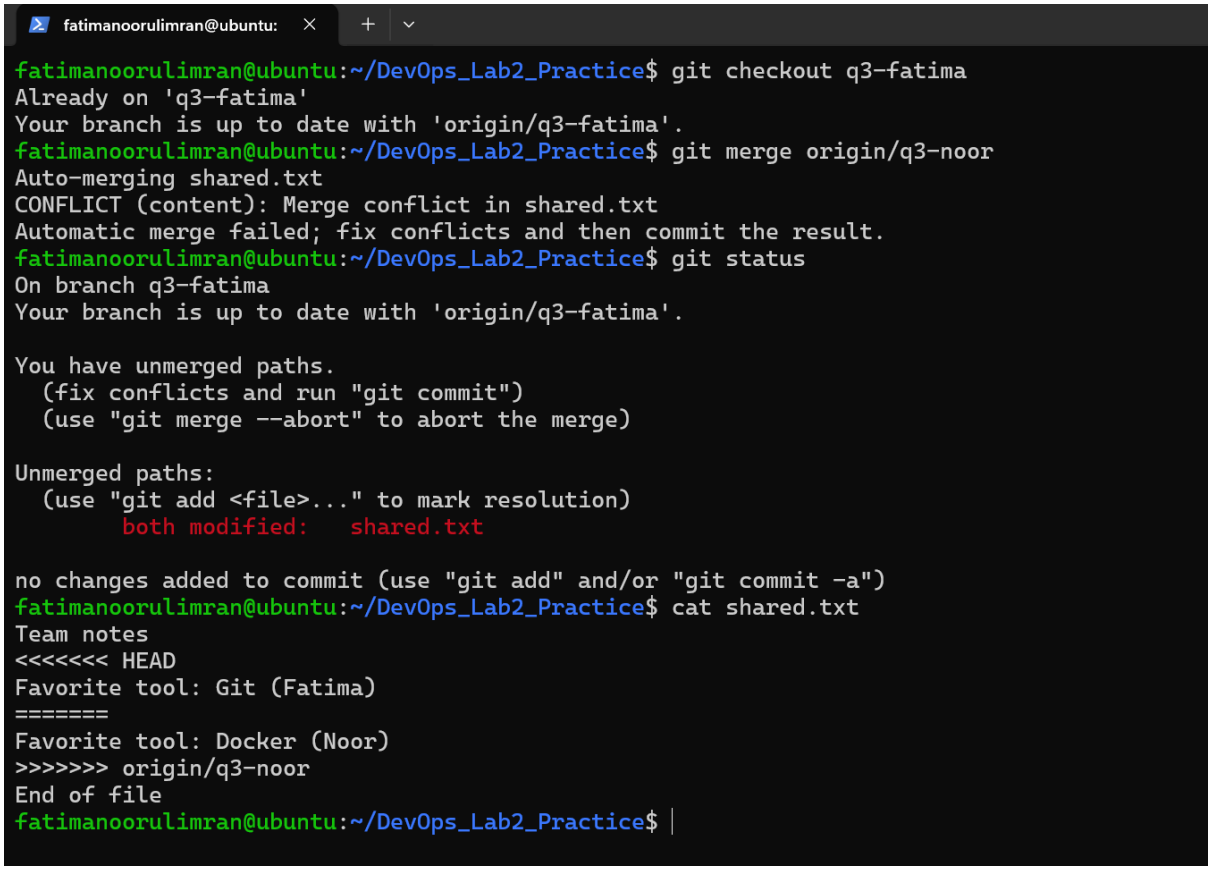
First contributor modified the file on their branch.



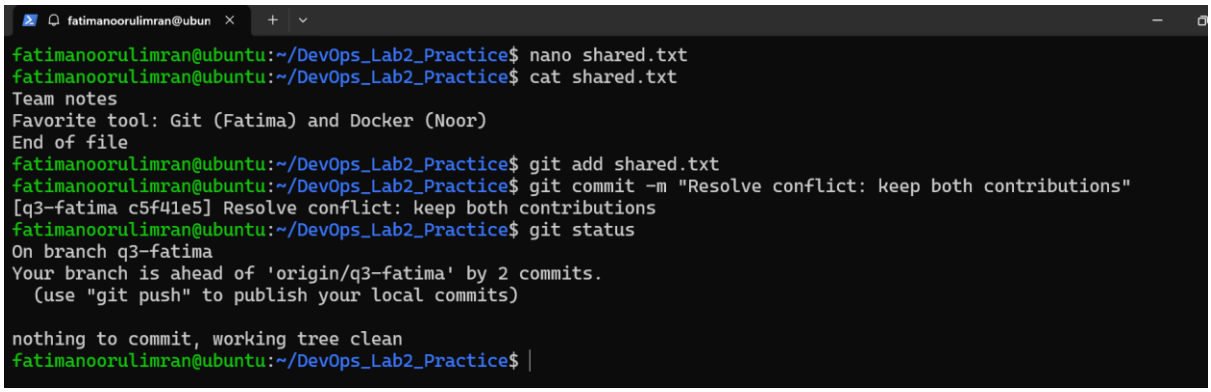
Second contributor modified the same file on a separate branch.



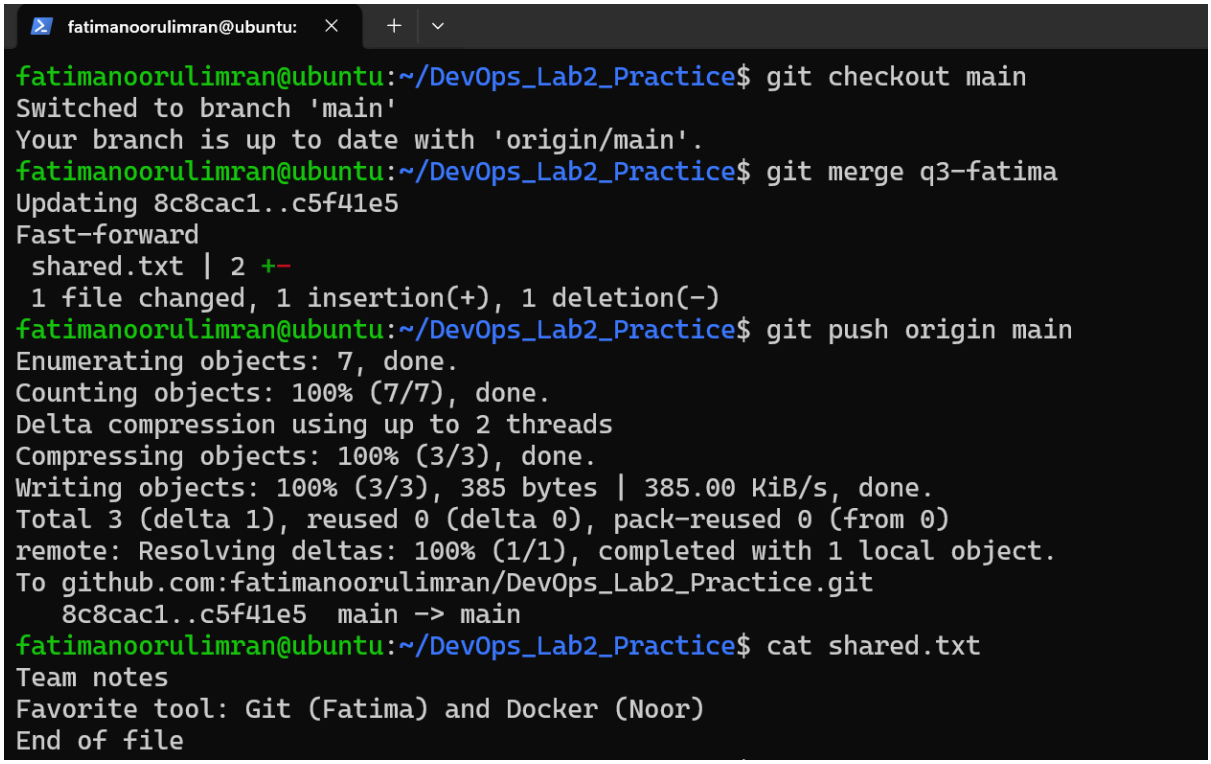
Both branches pushed to GitHub.



Merge conflict reported by Git.



Conflict resolved, keeping both contributions.



Final file containing both collaborators' changes.